

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

8 Data Science (djw1005)

Let $(E_0, E_1, \dots)$ be a Markov chain generated by

$$E_{t+1} = \lambda E_t + \text{Normal}(0, (1 - \lambda^2)\sigma^2)$$

where $0 \leq \lambda < 1$.

- (a) What is meant by “stationary distribution”? Show that $\text{Normal}(0, \sigma^2)$ is a stationary distribution for this Markov chain. [3 marks]
- (b) What is meant by “memorylessness”? Give an expression for the log likelihood of a sequence of values $(e_1, \dots, e_n)$, given the value for E_0 . [4 marks]

Klaus Hasselmann, who won the 2021 Nobel Prize, studied climate models in which short-term random fluctuations can have longer-term effects. Suppose we’re given a dataset $(y_0, y_1, \dots, y_n)$ of temperatures at timepoints $t = 0, 1, \dots, n$, and we use a Hasselmann-style model,

$$Y_t = \alpha + \beta \sin(2\pi\omega t) + \gamma t + E_t$$

where $(E_0, E_1, \dots)$ is a Markov chain as described above, and α, β , and γ are unknown parameters to be estimated.

- (c) Give an expression for the log likelihood of $(y_1, \dots, y_n)$, given the value for Y_0 . [Hint: First find the distribution of Y_{t+1} given Y_t .] [6 marks]
- (d) What is meant by a “linear model”? Write out a linear model that can be used to estimate the unknown parameters α, β , and γ (treating all other parameters as known). Identify the feature vectors. [7 marks]

Answer:

— *Solution notes* —

Part (a). A distribution π is a stationary distribution for a Markov chain $X_0, X_1, \dots$ if $X_0 \sim \pi$ implies that $X_t \sim \pi$ for all t . [The slightly simpler definition for a time-homogeneous Markov chain, which only requires $X_1 \sim \pi$, is fine.]

If $E_0 \sim N(0, \sigma^2)$ then

$$E_1 \sim \lambda N(0, \sigma^2) + N(0, (1 - \lambda^2)\sigma^2) \sim N(0, \sigma^2)$$

thus $N(0, \sigma^2)$ is a stationary distribution.

Part (b). A random process $X_0, X_1, \dots$ is memoryless if, for all t , X_{t+1} is conditionally independent of $(X_0, \dots, X_t)$ given X_t . Here, we're told the noise terms are a Markov chain, hence they're memoryless. So

$$\begin{aligned} \log \Pr(e_1, \dots, e_n | E_0 = e_0) &= \sum_{t=1}^n \log \Pr_{E_t}(e_t | E_{t-1} = e_{t-1}) \\ &= -\frac{n}{2} \log(2\pi\rho^2) - \frac{1}{2\rho^2} \sum_{t=1}^n (e_t - \lambda e_{t-1})^2 \end{aligned}$$

where $\rho = (1 - \lambda^2)\sigma^2$. (This is all like lecture notes section “Likelihood and memorylessness”).

Part (c). To simplify the algebra, let $\mu_t = \alpha + \beta \sin(2\pi\omega t) + \gamma t$ and $\rho^2 = (1 - \lambda^2)\sigma^2$. Then

$$\begin{aligned} Y_{t+1} &= \mu_{t+1} + E_{t+1} \\ &\sim \mu_{t+1} + \lambda E_t + N(0, \rho^2) \\ &= \mu_{t+1} + \lambda(Y_t - \mu_t) + N(0, \rho^2) \\ &= \mu_{t+1} - \lambda\mu_t + \lambda Y_t + N(0, \rho^2). \end{aligned}$$

So the log likelihood is

$$\log \prod_{t=1}^n \Pr_{Y_t}(y_t | Y_{t-1} = y_{t-1}) = -\frac{n}{2} \log(2\pi\rho^2) - \frac{1}{2\rho^2} \sum_{t=1}^n (y_t - (\mu_t - \lambda\mu_{t-1} + \lambda y_{t-1}))^2.$$

Part (d). A linear model is a supervised learning model in which the response is a linear combination of known features, weighted by unknown parameters. Typically we estimate the unknown parameters by finding the values that minimize the sum of squared errors.

To estimate the unknown parameters from the model from part (c): effectively we're finding a least-squares fit of the model

$$\begin{aligned} y_t &\approx \mu_t - \lambda\mu_{t-1} + \lambda y_{t-1} \\ &= \alpha + \beta \sin(2\pi\omega t) + \gamma t - \lambda\alpha - \lambda\beta \sin(2\pi\omega(t-1)) - \lambda\gamma(t-1) + \lambda y_{t-1} \\ y_t - \lambda y_{t-1} &\approx \alpha[1 - \lambda] + \beta[\sin(2\pi\omega t) - \lambda \sin(2\pi\omega(t-1))] + \gamma[t - \lambda(t-1)] \end{aligned}$$

This is a linear model, and the features are the terms in $[\cdot]$, and the vectors have one entry for each $t = 1, \dots, n$.
